Differential Geometry Notes

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September 2026

1 The derivative

Definition 1. *Define grad* $\in L(C^\infty(M), \mathcal{X}(M))$ *by* $\langle v, grad_f(p)\rangle = Df_p(v)$ *for all* $v \in T_pM$

Definition 2. *Define* $Hess_p \in L(C^\infty(M), T_pM^{(2)})$ *by*

$$f \to ((v, w) \to \langle \nabla_v X_f, w\rangle)$$

Claim 1. *range* $Hess_p \subseteq (T_pM)^{(2)}_{sym}$

Claim 2. *The map* $X, Y \to D_X Y$ *is bilinear and has the following properties:*

- $D_{fX}Y = f(D_X Y)$
- $D_X(fY) = f(D_X Y) + (D_X f)Y$
- $D_X Y - D_Y X = [X, Y]$
- $D_X\langle Y, Z\rangle = \langle D_X Y, Z\rangle + \langle Y, D_X Z\rangle$

Claim 3. *Suppose* $\phi : \Omega \to V$ *is a local parametrization. Then* $D_{\phi_i}\phi_j = \frac{\partial\phi}{\partial x_i \partial x_j}$

Suppose $F : M \to N$ is a diffeomorphism

Claim 4. *Suppose* $v \in T_pM$*. Then for all* $f \in C^\infty(M)$ *we have* $D_v(f \circ F) = (D_{F_*v})f$

Claim 5. *We have* $F_*[X, Y] = [F_*X, F_*Y]$

Claim 6. *We have* $\langle F_*X, F_*Y\rangle = \langle X, Y\rangle$

2 Riemannian metric

Definition 3. *Suppose* $\gamma : [a, b] \to M$ *is a piecewise smooth curve.*
A reparametrization of γ *is given by* $\gamma \circ \phi$*, where* $\phi : [c, d] \to [a, b]$ *is a homeomorphism with a partition* $c_0, c_1, \ldots, c_k$ *such that* $\phi_{|[c_{i-1}, c_i]}$ *is a diffeomorphism onto its image*

Claim 7. *For any γ piecewise regular, the map*

$$c \to length(\gamma_{[a,c]})$$

is a valid reparametrization

Claim 8. *Suppose M is connected. Then there is a piecewise regular curve between any two points*

Definition 4. *Suppose M is connected. Then we define the Riemannian metric by*

$$d_M(p,q) = inf\{L(\gamma) : \gamma \text{ is a piecewise smooth curve from } p \text{ to } q\}$$

Claim 9. *The topology induced by the Riemannian metric is the same as the subspace topology M inherits from $\mathbb{R}^k$*

Claim 10. *If there is a piecewise smooth curve from p to q of length $d_M(p,q)$ then there is such a piecewise regular curve as well. This is true for every p and q if M is closed in $\mathbb{R}^k$*

Definition 5. *A diffeomorphism $F : M \to N$ is called a Riemannian isometry if at each point the derivative DF_p is an isometry from T_pM to T_pN*

Claim 11. *Every Riemannian isometry is an isometry with respect to the Riemannian metric*

Definition 6. *A smooth map $F : M \to N$ is called a local Riemannian isometry at p if there is a neighborhood U such that $F(U)$ is open and $F_{|U} : U \to F(U)$ is a Riemannian isometry*

3 First fundamental form

Definition 7. *The first fundamental form of M at p is defined by restricting the standard inner product on $\mathbb{R}^k$ to the tangent space. We have $g_p : T_pM \times T_pM \to \mathbb{R}$*

Claim 12. *Two manifolds M and N admit a local Riemannian isometry from p to q if and only if there are local parametrizations $\phi : \Omega \to U$ and $\varphi : \Omega \to V$ such that $g_{\phi(a)}(v_i, v_j) = h_{\varphi(a)}(w_i, w_j)$ for all $a \in \Omega$, where $v_i = \phi_i(\phi(a))$ and $w_i = \varphi_i(\varphi(a))$*

4 Covariant derivative

Definition 8. *We have $\pi_p \in End(\mathbb{R}^k)$ which is the orthogonal projection onto T_pM. We define $\hat{\pi}_p$ as $I - \pi_p$*

Definition 9. *Suppose $v \in T_pM$. Define $\nabla_v \in L(\mathcal{X}(M), T_pM)$ by $X \to \pi_p(DX)_p v$*

Definition 10. *Suppose $X, Y \in \mathcal{X}(M)$ are smooth vector fields. Then we have the smooth vector field $\nabla_X Y = (p \to \pi_p(D_X Y)(p))$*

Claim 13. *The map $X, Y \to \nabla_X Y$ is bilinear and has the following properties:*

- $\nabla_X Y - \nabla_Y X = [X, Y]$
- $\nabla_{fX} Y = f(\nabla_X Y)$
- $\nabla_X (fY) = f(\nabla_X Y) + (D_X f)Y$

Definition 11. *Suppose $\phi : \Omega \to V$ is a local parametrization. The Christoffel symbols $\Gamma^k_{ij} \in C^\infty(\Omega)$ are defined by $(\nabla_{\phi_i}\phi_j)(\phi(a)) = \sum_k \Gamma^k_{ij}(a)\phi_k(a)$*

Definition 12. *The curvature tensor of X, Y, Z is defined by $R(X, Y, Z) = \nabla_X(\nabla_Y Z) - \nabla_Y(\nabla_X Z) - \nabla_{[X,Y]}Z$*

Suppose $F : M \to N$ is a Riemannian isometry

Claim 14. *We have $\nabla_{F_*X}(F_*Y) = F_*(\nabla_X Y)$*

Claim 15. *We have $\nabla_{F_*v} \circ F_* = DF_p \circ \nabla_v$*

Claim 16. *We have $F_*[R(X, Y, Z)] = R(F_*X, F_*Y, F_*Z)$*

5 Vector fields along a curve

Definition 13. *Suppose $\gamma : I \to M$ is a smooth curve. A smooth vector field along γ is a map $V : I \to \mathbb{R}^k$ such that $V(t) \in T_{\gamma(t)}M$*

The vector space of smooth vector fields along γ is $\mathcal{X}(\gamma)$

Definition 14. *We have $cov_\gamma \in L(\mathcal{X}(\gamma))$ given by $V \to (t \to \pi_{\gamma(t)}\dot{V}(t))$*

Definition 15. *A vector field V along γ is called parallel if $V \in ker(cov_\gamma)$*

Claim 17. *Suppose $t_0 \in I$, $p_0 = \gamma(t_0)$ and $v_0 \in T_{p_0}M$. There is a unique parallel vector field V satisfying $V(t_0) = v_0$*

Definition 16. *Let $t_1, t_2 \in I$. We have the parallel transport map $P_{t_1t_2} \in L(T_{\gamma(t_1)}M, T_{\gamma(t_2)}M)$*

Claim 18. *$P_{t_1t_2}$ is an isometry*

Claim 19. *$P_{t_1t_2}$ is invertible with inverse $P_{t_2t_1}$*

Claim 20. *Suppose t_0 is in the interior of I. Then we have*

$$\nabla_v X = lim_{t\to t_0}\frac{P_{tt_0}(X(\gamma(t))) - X_p}{t - t_0}$$

6 Second fundamental form

Definition 17. *N_pM is the orthogonal complement of T_pM in $\mathbb{R}^k$*

Definition 18. *The second fundamental form of M at p is the bilinear map $S_p : T_pM \times T_pM \to \mathbb{R}^k$ given by $v, w \to [(D\pi)_p v]w$*

Claim 21. *range $S_p \subseteq N_pM$ for all p*

Claim 22. *The second fundamental form is symmetric*

Claim 23. *We have $S_p(X_p, Y_p) = \hat{\pi}_p((D_X Y)(p))$*